

- 8 A long horizontal tube, containing fine powder, is closed at one end. A loudspeaker connected to a signal generator is positioned at the other of the tube. At a particular frequency, a stationary wave is set up inside the tube and the powder forms heaps as shown in Fig. 8.1.

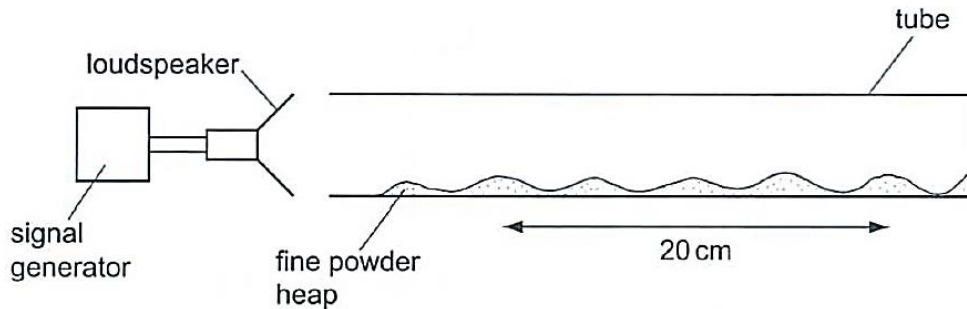


Fig. 8.1

The speed of sound is 330 m s^{-1} .

- (a) Explain why, for a stationary wave to form inside the tube, it is usually necessary to adjust either the frequency of the signal generator or the length of the tube.

[Turn over]

		$330 = f (10 \times 10^{-2})$ $f = 3300 \text{ Hz}$	C1 A1
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[Total: 6]

9	Read the passage below and answer the questions that follow
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